

Part 1: Summary of Lecture 1 (Data & Image Representation)

1. Data Representation Units:

- **Bit:** The smallest unit (0 or 1).
- **Nibble:** A group of **4 bits**.
- **Byte:** A group of 8 bits.
- **Word:** Typically composed of multiple bytes.

2. Image Representation (Raster Images):

- **Binary Image:** Also called **1-bit monochrome**. Contains no color (black/white only).
- **Gray Image:** Each pixel is **8 bits**. Values range from **0 to 255**.
 - *Example:* A 640x480 gray image requires **300 KB** of storage.
- **Color Image:** Each pixel is **24 bits** (RGB: Red, Green, Blue). Supports **16,777,216** colors.
 - *Example:* A 640x480 24-bit color image requires **921.6 KB** without compression.
- **7-Segments:** An image on 7-segments is represented by **7 bits**.

3. Sound & Video Representation:

- **Sound:** An analog acoustic wave.
 - **Amplitude:** Determines **Loudness**.
 - **Frequency:** Determines **Tone**.
 - **Sampling:** Typically **48kHz** per second.
- **Video:** A sequence of images called **Frames**. To create a continuous movie impression, frames are flashed faster than **50/sec**,.

4. Data Standards (Text):

- **ASCII:** American Standard Code for Information Interchange.
 - Uses 7 or 8 bits, limited to **256** combinations.
 - *Key Values:* '**A**' = **65**, '**a**' = **97**, **Space** = **32**.
- **Unicode:** Uses **2 bytes** (16 bits) per character.
 - *Advantage:* Represents **65,536** symbols.
 - *Disadvantage:* Takes **twice as much space** as ASCII.
- **UTF-8:** Variable length encoding (1 to 4 bytes).

- **Leading Byte:** Determines length (e.g., 110xxxxx means 2 bytes).
- **Continuation Byte:** Always starts with 10xxxxxx.
